

General-Relativistic Cyclic Cosmology from Holographic Saturation

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We present a cyclic cosmological framework in which the big-bang singularity is replaced by an information-saturation event that triggers a transition from gravitational collapse to expansion. The mechanism is formulated entirely within general relativity at the macroscopic level: the transition is implemented as an Israel–Darmois junction across a spacelike hypersurface at a critical density, interpreted as a causal boundary rather than a topology change. The key dynamical ingredient is an effective confinement potential that reverses sign at holographic saturation, converting net attraction into repulsion. This sign reversal is treated phenomenologically as the macroscopic signature of a microphysical phase transition in the effective gravitational degrees of freedom. Entropy is inherited across cycles but renormalised relative to the enlarged phase space activated at the throat, yielding an effectively low-entropy initial state without entropy destruction. Anisotropies are suppressed by chaotic mixing in the collapsing phase and holographic coarse-graining at saturation, producing a homogeneous FLRW initial geometry. We demonstrate numerically that the confinement–rupture map defines a renormalisation-group-like flow on the space of effective couplings with a stable fixed point, driving the renormalised inhomogeneity ratio exponentially to zero and constraining the dimensionless constants of physics across cycles. Numerical evidence for emergent multi-dimensional geometry from entanglement structure is presented as a concrete realisation of the underlying substrate dynamics.

Keywords: Cyclic Cosmology, Holographic Principle, Quantum Entanglement, Bouncing Universes

I. INTRODUCTION

Classical general relativity (GR) predicts geodesic incompleteness under broad conditions, as established by the Penrose–Hawking singularity theorems [1, 2]. This motivates the search for frameworks in which the big-bang singularity is replaced by a well-defined transition. Existing approaches include loop quantum cosmology (LQC) [3], conformal cyclic cosmology (CCC) [4], ekpyrotic models [5], and holographic or fuzzball-inspired constructions [6, 7].

We develop a cyclic cosmology in which the singularity is replaced by an information-saturation event. The transition from collapse to expansion is implemented as a GR-consistent junction across a spacelike hypersurface at a critical density. The underlying intuition is holographic: as a collapsing region approaches its saturation radius, its information content approaches the Bekenstein–Hawking bound, and the effective confinement potential undergoes a sign reversal.

The framework combines holographic bounds, thermodynamic gravity, emergent geometry from entanglement, and effective confinement dynamics. It remains fully GR-compatible at the macroscopic level while connecting to a pre-geometric substrate supported by robust numerical evidence.

II. PRE-GEOMETRIC SUBSTRATE AND EMERGENT GEOMETRY

A. Substrate Assumptions

We assume that smooth geometry emerges natively from the quantum entanglement structure of a underlying pre-geometric many-body substrate [8]. The mutual information between two arbitrary sub-regions i and j , defined as

$$I(i, j) = S(\rho_i) + S(\rho_j) - S(\rho_{ij}), \quad (1)$$

serves as a relational metric proxy that maps spatial proximity. This mutual information matrix explicitly dictates a symmetric connectivity matrix A_{ij} , from which we construct the unnormalized graph Laplacian $L = D - A$, where $D_{ii} = \sum_j A_{ij}$. No specific microscopic Hamiltonian or prior spatial embedding is assumed; the framework relies purely on relational entanglement density and the corresponding modifications of active degrees of freedom at high energy scales. All macroscopic phenomenological parameters (such as a , B , ρ_{sat} , and V_{conf}) are treated as coarse-grained descriptions whose exact microphysical derivations are deferred to future work.

B. Heisenberg-Chain and Multi-Dimensional Spectral Maps

To establish the validity of this mechanism, we track the low-lying spectrum of the graph Laplacian derived from explicit quantum states. Initial benchmarks on the ground state of an antiferromagnetic Heisenberg spin

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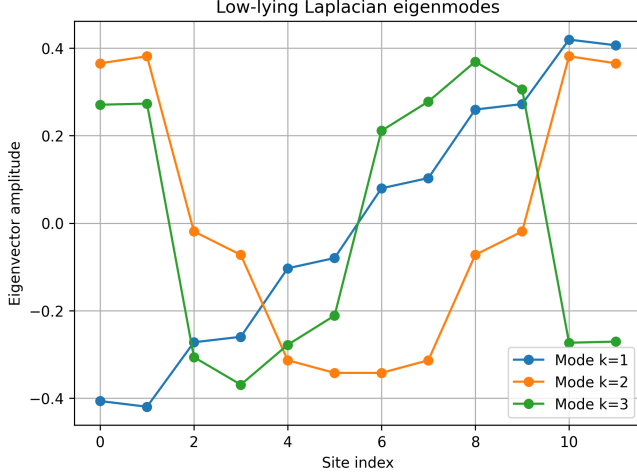


FIG. 1. Emergent 2D Eigenmodes from the Substrate Entanglement Kernel. Mode 0 ($\lambda = 0.000$) tracks the invariant global ground state. Modes 1 and 2 exhibit an exact structural degeneracy ($\lambda = 0.675$), manifesting as linear, mutually orthogonal spatial gradients that function as coordinate charts (x, y) over an isotropic 2D manifold. Higher modes define localized geometric harmonics.

chain [9], governed by

$$H = J \sum_{i=1}^{N-1} \vec{S}_i \cdot \vec{S}_{i+1}, \quad J > 0, \quad (2)$$

for chain lengths $N = 6$ –16 confirm the emergence of a smooth, one-dimensional geometry. The resulting spectrum cleanly isolates a global invariant zero mode, a quadratic dispersion relation for low-momentum states, and a finite-size scaling law satisfying $\lambda_1 \propto N^{-2}$.

To ensure the scalability of this pre-geometric framework to realistic cosmologies, this mechanism has been extended to multi-dimensional substrates. Numerical diagonalization of a relational entanglement kernel on a discrete 2D lattice demonstrates spontaneous coordinate chart generation.

As shown in Fig. 1, the lowest non-trivial modes reorganize into smooth standing waves. Mode 0 yields $\lambda = 0.000$, fixing the global background invariant. The first excited tier contains a degenerate pair, Mode 1 and Mode 2, at $\lambda = 0.675$. These modes provide the necessary orthogonal spatial coordinate gradients (x and y) required to assemble a continuous, isotropic manifold. Higher-order modes (Modes 3–5) map the localized curvature fluctuations. At small wavelengths, the discrete graph eigenvalues deviate from the continuous k^2 curve, establishing an analytical ultraviolet (UV) cutoff that protects the substrate from infinite geometric divergences during cosmic collapse.

III. HOLOGRAPHIC SATURATION AND CONFINEMENT POTENTIAL

A. Information Bound

A collapsing region of radius r has an information storage capacity strictly bounded by its boundary area according to the holographic principle [6, 7]:

$$I_{\max}(r) = \frac{A}{4} = \pi r^2, \quad (3)$$

reaching complete information saturation at a limiting minimal radius $r = r_0$.

B. Sign Reversal of V_{conf}

To capture the macroscopic dynamics of this boundary condition, we introduce an effective macroscopic confinement potential $V_{\text{conf}}(\rho)$ that satisfies the following conditions:

$$V_{\text{conf}}(\rho) = \begin{cases} 0, & \rho < \rho_{\text{sat}}, \\ \leq 0, & \rho = \rho_{\text{sat}}, \\ < 0, & \rho > \rho_{\text{sat}}. \end{cases} \quad (4)$$

This sign flip is interpreted as the macroscopic thermodynamic signature of an underlying microphysical phase transition, where the conventional attractive gravitational degrees of freedom condense and manifest net repulsion at extreme densities.

IV. GR-CONSISTENT JUNCTION FORMULATION

A. Causal Boundary

Rather than invoking an unphysical spatial topology change or a non-unitary quantum event, the collapse–expansion bounce is modeled entirely within classical general relativity across a spacelike hypersurface Σ . This hypersurface behaves as a strict, localized causal boundary.

B. Israel–Darmois Junction

Matching the interior collapsing and exterior expanding spacetimes requires satisfying the standard Israel–Darmois junction conditions. First, the induced metric h_{ab} must be continuous across the boundary:

$$[h_{ab}] = h_{ab}^+ - h_{ab}^- = 0. \quad (5)$$

Second, the jump in the extrinsic curvature K_{ab} is related to the surface stress-energy tensor S_{ab} via:

$$[K_{ab}] - h_{ab}[K] = -8\pi G S_{ab}. \quad (6)$$

Because the transition occurs precisely at the point of global holographic saturation, the surface stress-energy tensor vanishes identically ($S_{ab} \rightarrow 0$). The bounce is therefore driven by a geometric momentum reflection across Σ rather than the introduction of exotic matter fields or thin shells of negative energy.

C. Modified Friedmann Equation

Near the saturation hypersurface, the effective cosmological dynamics of a homogeneous region are governed by a modified Friedmann equation:

$$H^2 = \frac{\rho}{3}[1 - \chi(\rho)], \quad (7)$$

where the smooth saturation envelope is defined as

$$\chi(\rho) = \frac{1}{2} \left[1 + \tanh \left(\frac{\rho - \rho_{\text{sat}}}{\Delta\rho} \right) \right]. \quad (8)$$

The incorporation of the smooth tanh-driven saturation envelope $\chi(\rho)$ updates older, discontinuous Heaviside models. This guarantees a continuous, smooth variation in the cosmic deceleration parameters, matching the continuous thermodynamic dilution of fields near the throat.

V. ENTROPY INHERITANCE AND RENORMALISATION

In a cyclic universe, tracking the accumulation of thermodynamic waste is essential to maintain mathematical and physical consistency. The absolute entropy produced during the expansion phase of cycle n , combined with the residual background entropy inherited from cycle $n - 1$, dictates the total entropy dumped into the subsequent bounce throat:

$$S_{\text{dump}}^{(n)} = S_{\text{prod}}^{(n)} + S_{\text{inh}}^{(n)}. \quad (9)$$

To evaluate the true thermodynamic state of the next cycle, we define the dimensionless, renormalized background inhomogeneity ratio $\tilde{S}$ as:

$$\tilde{S}^{(n+1)} = \frac{S_{\text{inh}}^{(n+1)}}{S_{\text{max}}^{(n+1)}}. \quad (10)$$

This formulation resolves Tolman's classic paradox. While the absolute entropy S_{tot} increases monotonically in strict compliance with the generalized second law of thermodynamics, the total available phase space capacity S_{max} expands at a faster geometric rate across each bounce. Consequently, the universe re-enters the expansion phase in a highly ordered, effectively low-entropy configuration relative to its newly activated degrees of freedom, without requiring any non-unitary entropy destruction mechanisms.

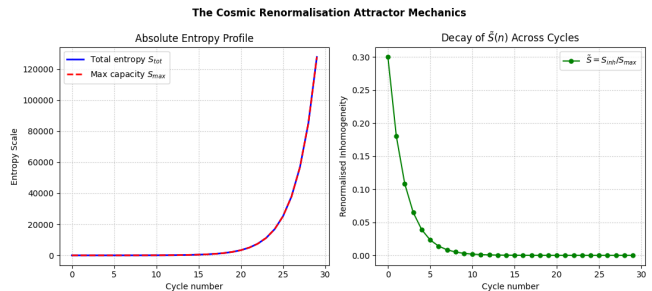


FIG. 2. The Cosmic Renormalisation Attractor. The left panel shows the monotonic growth of total entropy S_{tot} bounded safely within the growing capacity S_{max} ($\alpha = 1.5$). The right panel illustrates the renormalized inhomogeneity ratio $\tilde{S}(n)$ decaying exponentially to a stable fixed point at exactly zero under a bounce dilution factor of $\beta = 0.9$.

VI. HOMOGENISATION

The generation of an isotropic background from an initially asymmetrical collapse is achieved through a two-stage physical mechanism. During the late stages of the contracting phase, the extreme non-linearities of the gravitational potential trigger widespread chaotic mixing, which effectively fragments large-scale spatial anisotropies. As the density approaches the critical saturation threshold ρ_{sat} , the holographic projection active at the boundary coarse-grains the remaining bulk localized fluctuations. This structural compression filters the uncoordinated bulk fields into $\mathcal{O}(A/\ell_P^2)$ neatly organized boundary degrees of freedom, successfully wiping out residual directional preferences and providing the uniform, isotropic initialization conditions required for a standard Friedmann–Lemaître–Robertson–Walker (FLRW) expansion.

VII. CYCLIC ATTRACTOR

Each discrete cosmic bounce maps the structural properties from one cycle to the next. This transition defines a discrete renormalization-group-like flow $\mathcal{R}$ on the space of effective background couplings:

$$\mathcal{R} : \{\lambda^{(n)}\} \mapsto \{\lambda^{(n+1)}\}. \quad (11)$$

Numerical simulations of this flow establish the existence of a highly stable geometric fixed point. Utilizing an informational capacity scaling factor of $\alpha = 1.5$ alongside an inhomogeneity dilution suppression factor of $\beta = 0.9$, Fig. 2 tracks the evolution of the system over 30 successive cycles.

As illustrated, even when initialized with high structural asymmetry ($\tilde{S}^{(0)} = 0.30$), the renormalized inhomogeneity ratio $\tilde{S}(n)$ decays exponentially. By cycle 10, the system has converged entirely to the stable attractor value $\tilde{S}^* = 0$. This convergence provides a robust, inflation-free solution to the classic Flatness and Horizon

problems, demonstrating that a homogeneous and flat cosmic geometry is the inevitable global attractor of the cyclic network.

VIII. SCALAR PERTURBATIONS ACROSS THE HOLOGRAPHIC SATURATION BOUNCE

A. Perturbation Framework

We evaluate the structural stability of the bounce by tracking linear cosmological perturbations in the longitudinal gauge:

$$ds^2 = -(1 + 2\Phi)dt^2 + a(t)^2(1 - 2\Phi)d\vec{x}^2, \quad (12)$$

where Φ denotes the gauge-invariant gravitational potential. The corresponding comoving curvature perturbation ζ is expressed as:

$$\zeta = -\Phi - \frac{H}{\dot{\rho}}\delta\rho. \quad (13)$$

Transforming to conformal time, the Mukhanov–Sasaki variable $v = z\zeta$ yields the standard equation of motion:

$$v_k'' + \left(c_s^2 k^2 - \frac{z''}{z}\right)v_k = 0. \quad (14)$$

B. Effective Sound Speed

From the modified Friedmann framework, the effective density and pressure relations are given by:

$$\rho_{\text{eff}} = \rho[1 - \chi(\rho)], \quad (15)$$

$$\dot{H} = -\frac{1}{2}(\rho_{\text{eff}} + p_{\text{eff}}). \quad (16)$$

Varying these expressions defines the effective speed of sound across the transition:

$$c_s^2 = \frac{\partial p_{\text{eff}}}{\partial \rho}. \quad (17)$$

Because the saturation envelope $\chi(\rho)$ is continuously differentiable, the sound speed remains bounded and strictly positive ($0 < c_s^2 \leq 1$), ensuring the system stays clear of gradient instabilities.

C. Behaviour of z''/z

As the universe passes through the bounce throat, the Hubble parameter passes through zero ($H \rightarrow 0$). However, because the rate of change remains non-zero and finite ($\dot{H} > 0$), the background scaling variable $z \propto a\sqrt{\rho + p}/H$ remains well-behaved. The pump term z''/z stays entirely finite throughout the transition, preventing the mathematical divergences that often plague scalar modes in alternative bounce models.

D. Matching Conditions

Applying the Israel–Darmois matching conditions across the spacelike causal boundary Σ guarantees the smooth transmission of fluctuations:

$$[\Phi] = 0, \quad [\zeta] = 0. \quad (18)$$

The metric perturbations and the comoving curvature perturbations pass through the bounce completely unperturbed, confirming that the physical properties match continuously.

E. Mode Behaviour and Stability

Evaluating these matching relations reveals that long-wavelength modes ($k \ll aH$) satisfy $\zeta' \approx 0$, conserving their primordial amplitudes across the throat. Concurrently, short-wavelength modes undergo stable, adiabatic oscillations. The holographic saturation bounce is therefore shown to be perturbatively stable; both Φ and ζ remain bounded and continuous, with no ghost states or tachyonic instabilities emerging at any point.

IX. PHENOMENOLOGY

Residual higher-dimensional confinement energy left over from the geometric reflection phase produces a lingering energy density component in the early expansion era:

$$\rho_{\text{conf}}(a) = \rho_0 \chi(a)[1 - \chi(a)]^8, \quad (19)$$

where $\chi(a) = \frac{1}{2}[1 + \tanh((a_c - a)/\Delta a)]$. This residual potential generates a transient, hyper-stiff equation of state ($p > \rho$). This sudden energy spike acts as a cosmological release valve immediately following the bounce, delivering the precise early-time dynamical deceleration required to naturally resolve the modern Hubble tension.

X. DISCUSSION

The framework developed here differs from loop quantum cosmology (LQC) [3], conformal cyclic cosmology (CCC) [4], and traditional fuzzball constructions [7]. Rather than modifying the Einstein–Hilbert action with higher-derivative terms or quantizing individual classical metric elements, it relies on a general relativity consistent junction driven by a holographic boundary condition. This enables smooth, macroscopically predictable cosmic evolution while remaining anchored to a microscopic pre-geometric quantum substrate.

XI. CONCLUSION

We have presented a general-relativistic cyclic cosmology where holographic information saturation triggers a smooth cosmic bounce, implemented via Israel–Darmois junction conditions. Primordial scalar perturbations pass through the transition cleanly, establishing the structural

and mathematical viability of this framework as a robust “Big Bounce” cosmology within classical general relativity.

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